

# DATA-260 Homework 4 Report

## Student Information

- Student: Shruti Shetty
- SID4: 9004
- Domain: Open-Source Package Vulnerabilities
- Branch: hw4
- Database: s9004\_rel
- Seed: 9004
- Verify seed: 269004

## 1. Project Overview

This project extends the HW3 vulnerability-report application.

The HW4 system includes:

- A React frontend
- FastAPI backend routes
- MySQL persistence
- HTTP-only cookie authentication
- Server-side sessions
- Create, read, update, and delete operations
- N+1 query benchmarking
- RAG document retrieval and evaluation

## 2. React Frontend

The React client provides login, report listing, report creation, report updating, and report deletion.

Main React components:

- `Login.jsx`
- `Home.jsx`
- `CreateRecord.jsx`
- `UpdateRecord.jsx`
- `DeleteRecord.jsx`

The frontend uses Axios to communicate with FastAPI and React Router for navigation.

Important routes:

- /
- /create
- /update
- /delete

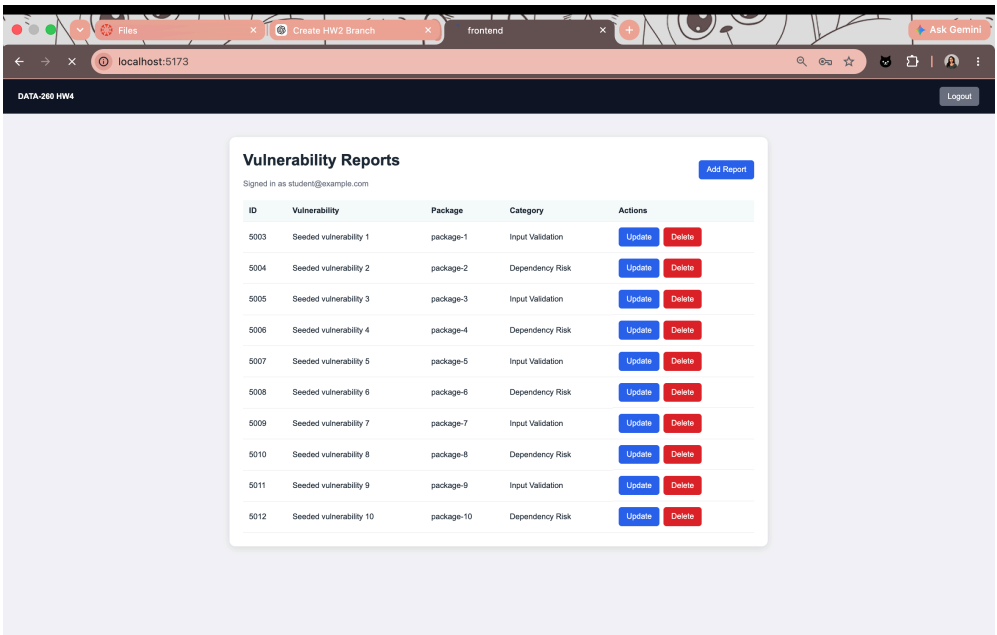
The frontend was tested by logging in, creating a report, viewing it, updating it, deleting it, and logging out.

Example API call:

```
```javascript await api.post("/api/reports", payload) ## 10. Evidence Screenshots
```

## React Dashboard

The React dashboard displays the logged-in user and seeded vulnerability records with Update and Delete actions.



React dashboard

## HW4 Verification

The verification output confirms the database, N+1 benchmark, RAG, and frontend checks.

```
~/data260-9004 -- zsh                                ~/data260-9004 -- zsh                                ~/data260-9004/nrmcmd -- zsh                                ...260-9004 -- curl http://localhost:8004/health                                +

{"model": "sentence-transformers/all-MiniLM-L6-v2",
 "database": "s9004_rel",
 "checks": {
   "corpus_has_at_least_5_documents": true,
   "database_has_8000_vulnerabilities": true,
   "database_has_200_related_items": true,
   "n_plus_one_raw_json_valid": true,
   "n_plus_one_has_100_requests": true,
   "n_plus_one_metrics_valid": true,
   "n_plus_one_has_6_metric_groups": true,
   "rag_results_valid": true,
   "rag_evaluation_valid": true,
   "rag_k_sweep_valid": true,
   "rag_printouts_present": true,
   "react_api_present": true,
   "react_app_present": true,
   "database_modelis_present": true,
   "benchmark_script_present": true,
   "rag_script_present": true,
   "run_log_present": false,
   "metrics_present": false,
   "ai_use_present": false,
   "report_pdf_present": false
 },
 "status": "incomplete"
}
(base) srusti@Aashiks-MacBook-Pro data260-9004 % cd ~/data260-9004
code reports/nw04/RUN_LOG.txt
(base) srusti@Aashiks-MacBook-Pro data260-9004 % cd ~/data260-9004
code reports/nw04/METRICS.md
(base) srusti@Aashiks-MacBook-Pro data260-9004 % cd ~/data260-9004
code reports/nw04/AI_USE.md
(base) srusti@Aashiks-MacBook-Pro data260-9004 % cd ~/data260-9004
touch reports/nw04/report.md
code reports/nw04/report.md
(base) srusti@Aashiks-MacBook-Pro data260-9004 % uvicorn api:app --app-dir code --reload --port 8004
zsh: command not found: uvicorn
(base) srusti@Aashiks-MacBook-Pro data260-9004 % (base) srusti@Aashiks-MacBook-Pro data260-9004 % uvicorn api:app --app-dir code --reload --port 8004
zsh: command not found: uvicorn
(base) srusti@Aashiks-MacBook-Pro data260-9004 %
zsh: parse error near 'srusti@Aashiks-MacBo...'
(base) srusti@Aashiks-MacBook-Pro data260-9004 % uvicorn api:app --app-dir code --reload --port 8004
zsh: command not found: uvicorn
(base) srusti@Aashiks-MacBook-Pro data260-9004 % conda activate data260
(data260) srusti@Aashiks-MacBook-Pro data260-9004 % uvicorn api:app --app-dir code --reload --port 8004
INFO: Will watch for changes in these directories: ['/Users/srusti/data260-9004']
ERROR: [Errno 48] Address already in use
(data260) srusti@Aashiks-MacBook-Pro data260-9004 % uvicorn api:app --app-dir code --reload --port 8004
INFO: Will watch for changes in these directories: ['/Users/srusti/data260-9004']
ERROR: [Errno 48] Address already in use
(data260) srusti@Aashiks-MacBook-Pro data260-9004 % ls -l :8004
COMMAND PID USER FD TYPE DEVICE SIZE/OFF NODE NAME
python3.1 26564 srusti 3u IPv4 0x559eb6c2e8a692e3 0t0 TCP localhost:8004 (LISTEN)
python3.1 98951 srusti 3u IPv4 0x559eb6c2e8a692e3 0t0 TCP localhost:8004 (LISTEN)
(data260) srusti@Aashiks-MacBook-Pro data260-9004 % curl http://localhost:8004/health
```

## HW4 verification